

Traditional VPNs work on the network layer, usually Layer 3 or Layer 4. They create an encrypted tunnel that ends right at the corporate network's edge. Once a user has authenticated, their machine is assigned an internal IP address. From the QA perspective, this means that the client has full access to all internal IP ranges and subnets. If I were to run a network scan using tools like Nmap in such a VPN, I will be able to see the various internal hosts and check for any network layer vulnerabilities. However, finding the traffic for a single web application becomes messy because the packet captures will also pick up background network noise, ARP requests, and broadcast traffic, leading to a very cluttered view.

In contrast to the VPN, ZTNA, like InstaSafe, operates strictly on the Application Layer which is Layer 7. The secure tunnel doesn't terminate at the network edge, but ends at the ZTNA gateway. The client is never given an internal IP address and has zero visibility into the underlying network infrastructure. For QA testing, this makes a highly controllable environment. Network level vulnerability discovery is impossible at this time because the client cannot see any internal routing at all. Instead, QA has to focus fully on identity, attached API boundaries, intercepting HTTP headers, validating the JWT tokens in browser DevTools, and checking micro-segmented application logic without using ping.